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Design and Implementation of Computer Integrated Manufacturing in Small and Medium-Sized Enterprises: A Case Study

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In recent years, there has been an increasing interest in the implementation of computer integrated manufacturing (CIM). Since many manufacturing organisations compete in international markets, a company strategy should be justified before making any major advanced technology-related investment decisions. The bottleneck for realising CIM systems in small and medium-sized enterprises (SMEs) is often the lack of inhouse knowledge about how to design a CIM system and how to implement the existing technological components to improve the manufacturing performance. It is necessary to develop a framework for the design and implementation of CIM in SMEs. The literature on the design and implementation of CIM is reviewed, a framework for the design and implementation of CIM in SMEs is developed, and a case study is conducted which involves collecting and analysing data from a small company, Company A, to develop a practical framework for the implementation of CIM in SMEs.*

Keywords: Case study; CIM; Implementation; SMEs

1. Introduction

In general, small and medium-sized enterprises (SMEs) are usually classified as enterprises carrying out all forms of small to medium-scale manufacturing. These are establishments with fewer than 500 employees and an annual turnover of £20 million. For all enterprises, the long-term goal is to stay in business, grow, and make a profit. This is especially so for manufacturing enterprises, which must understand the dynamic changes that are taking place in the business environment. The twenty-first century business environment can be characterised by expanding global competition and products of increasing variety and lower demand [1]. The globalisation of economic activity has brought about a change in the attitudes of customers. Today, customers take both minimum cost and high quality for granted. However, factors such as delivery performance and customisation, and environmental issues such as waste generation, are assuming a predominant role in defining the success of organisations in terms of increased market share and profitability [2]. There have been a number of reviews of the factors which influence the increasing importance of SMEs. SMEs generally have different markets from large firms. They generally manufacture more for the domestic markets, of items such as consumption goods, investment goods and suppliers' products and services, whereas large firms manufacture mainly for foreign markets. The business cycle shows synchronous fluctuations in both the domestic and foreign markets. Hence, because of their divergent market orientations, small and large firms experience different cyclical effects on production [3]. The cost structure of small enterprises is another reason why cyclical influences can affect SMEs differently from large enterprises. SME production is characterised by relatively high wage costs and profit margins, but they spend less than large firms on raw materials, semi-finished products and other products and services supplied to them. Whatever strategies are adopted by SMEs depend also on the economic and structural changes occurring in the economies in which they operate, and whatever type of strategy is implemented preserves the SME's flexibility. Small autonomous entrepreneurs having relatively low capital costs and mostly operating in highly competitive markets, can be more flexible with time-consuming internal consultations. They

*For commercial reasons, the identity of the company is concealed.

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are more able to respond to opportunities in a contracting market, and can adjust their prices to reflect cost changes [4]. Furthermore, lower overhead costs enable small entrepreneurs to implement price reductions earlier than large companies, without immediately risking serious losses.

Computer integrated manufacturing (CIM) is a label for a set of techniques that are making fundamental changes to manufacturing industry. CIM is a multidisciplinary subject and is a complex, multilayered system designed to minimise waste and create wealth in the broadest sense. According to Lefebvre et al. [5], CIM is concerned with providing computer assistance, control, and high-level integrated automation at all levels of manufacturing (and other) industries, by linking islands of automation into a distributed processing system. These isolated automated production islands include NC machines, distributed numerical control (DNC), computerised numerical control (CNC), material requirement planning (MRP), manufacturing resource planning (MRP II), CAD, computer-aided process planning (CAPP), computer-aided manufacturing (CAM), automated storage, computer-controlled material-handling equipment, and robotics.

For a manufacturing organisation to remain competitive, it must deliver products to customers at the minimum possible cost, at the best possible quality, and in the minimum lead time, starting from the product conceptual stage to final delivery, service, and disposal. The ultimate aim of CIM is to produce the correct number of parts of acceptable quality at the right time. If CIM technologies are fully integrated, the manufacturer can respond rapidly to changes in product design, demand, or product mix. It can effectively serve both small customers requiring only a few parts and major customers requiring large quantities of many different parts. The mix can be made with consistent quality and with minimum waste. Work in progress (WIP) and unscheduled downtimes can be reduced considerably. A high level of integration is required in all these activities. CIM is the concept of a totally automated factory in which all of the above elements are controlled by computers in an integrated manner [6].

The high investment required for CIM implementation is becoming a major hurdle for SMEs. Ten years ago, the majority of SMEs regarded CIM as only remotely relevant. The perception was that only the larger enterprises were affected and that only they could benefit from using more advanced technologies and from adapting new matching organisational principles [7]. The lack of financial resources has slowed the adoption of CIM. However, over the last decade, it has become clear that to bridge the gap between SMEs and large enterprises, the competitive advantage of CIM technologies is necessary. The management of technology is a challenge for all types of companies [8].

Research on the adoption of CIM technologies by SMEs has increased in recent years, including both “hard” technologies (e.g. robotics) and “soft” technologies (such as CAD). This seamless integration provides SMEs with easy communications both within its organisation and with suppliers, which enables them to improve their potential in the global market [9]. Many SMEs in the past have automated various operations independently within their organisations. These islands of automation have improved the local productivity, but are not sufficient to provide the necessary logistic support to improve productivity and quality of the whole organisation [10]. Manufacturers are beginning to realise that it is vital to have a structured approach to the introduction of this technology.

According to Kochan and Cowan [11], companies that have already adopted different sorts of CIM technologies are finding it difficult to put them into practice because of the threads linking different IT products. Therefore, it is imperative to formulate a strategy for implementing CIM at the earliest possible stage. There is no standard way for a company to adopt computers to create a CIM plant. All companies operate in different ways and in different market places, and each one must devise its own strategy for success. Hence, a company must first analyse its business and determine a priority of business objectives before it can start to draw up a plan for CIM. A good quality strategy will have the robustness to respond to any change of priorities [12].

This paper presents a brief background of CIM and SMEs along with past experiences of CIM in SMEs. A framework for the design and implementation of CIM in SMEs has also been discussed. The details of a case study conducted with a small company are presented. Finally, a summary of findings and recommendations along with conclusions are presented.

2. A Framework for the Implementation of CIM in SMEs

Based on the review of the characteristics of SMEs and large enterprises, and reported frameworks and case studies of the design and implementation of CIM, a conceptual framework for the design and implementation of CIM in SMEs has been developed, as shown in Fig. 1. A fundamental objective of CIM is optimisation of command, control and communication. There is inevitably a wide range of social, organisational and managerial issues linked with engineering and manufacturing automation. Information processing is an important factor for achieving communication within the organisation, and effective information management is a major priority in successful manufacturing systems.

This conceptual model is a development sequence in three stages for a CIM system in SMEs.

1. Current situation analysis.
2. Design.
3. Implementation.

The current situation analysis shows the company's adoption of CIM technology with reference to the theoretical framework [6]. It carries out and evaluates an analysis of the operational information, with the existing CIM technologies defined. The results should reflect the bottlenecks identified in the organisation of any CIM process. At the design stage, the aims and benefits of implementing CIM are defined with consideration of the business and manufacturing strategies, which have the shortest throughput time or the better capacity utilisation. Any rectification is then planned together with the necessary technical architecture, resulting in an agreed level of CIM complexity. Organisational behaviour is also considered to prepare for staff training or to employ the necessary expertise at this stage. The organisational prerequisite is the formation of a project team, which accompanies the project from planning through to its concluding integration. This team has the task of controlling the development process, in line with the overriding objectives and strategies.

Finally, in the step by step, time critical, implementation stage, the range of tasks should be carried out incrementally, and extend from the technological sector via information technology to commercial issues. For an SME, it is of paramount importance to buy in CIM modules which are available on the market and which are capable of being installed in the company and being compatible with the specifications of other systems installed or to be installed. This is to reduce the amount of modification that might have to be maid due owing to the diversity of requirements in different companies. Once the system has been integrated, it should simulate all organisational aspects in an SME.

3. A Case Study. Company A

In this section, the details of a case study are presented. The framework discussed (see Fig. 1) in the previous section has been used to analyse the status of Company A (Co. A) with reference to the implementation of CIM. This study is based on the data collected from the case company on the implementation of CIM with the help of structured interviews with managers using questionnaires. The details of the current system, the design of the CIM system, and the implementation are presented here.

Company A is a medium-sized company which has about 200 employees with an annual turnover of £8m. Reporting to the managing director, there is a finance director, technical director, production director, quality manager and marketing director. The production director and technical director are responsible for a large part of production operation in the company. Reporting to the production director, the manufacturing processes manager has responsibility for the

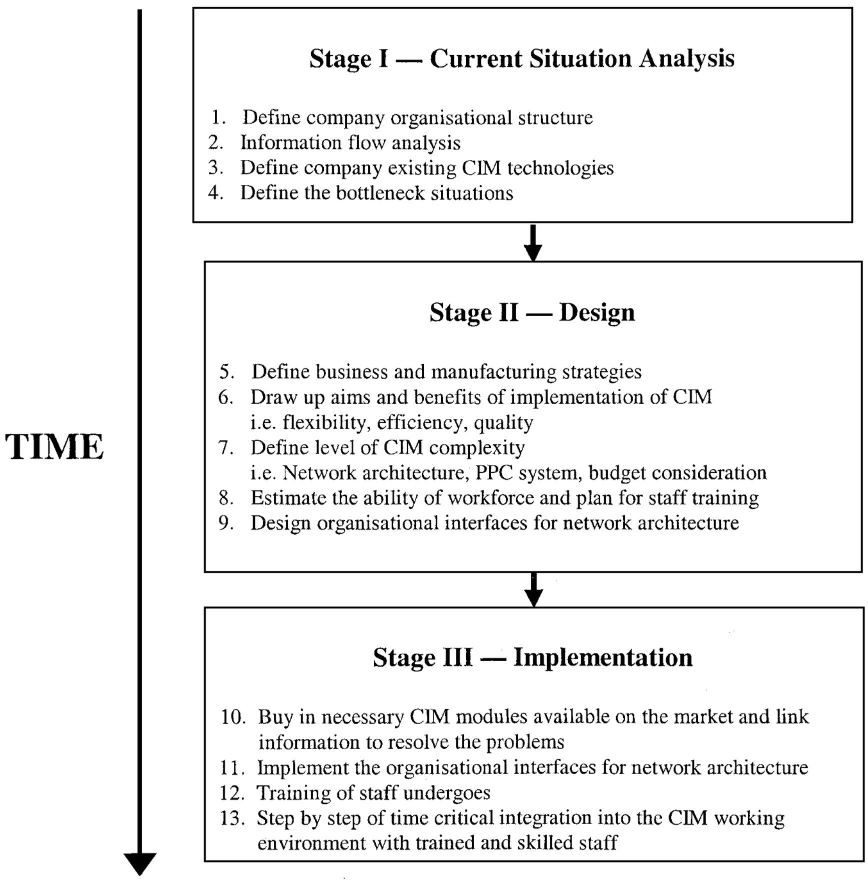

Fig. 1. A framework of the design and implementation of CIM in SMEs.

MRP system which Co. A regards as its main CIM technology. The main products of the company are small electronic devices with wide applications in many engineering industries. The company products have technical advantages over competing devices which are made by a few competitors in Europe and the far East. Its biggest competitor is a Japanese manufacturer. Since 1990, prices for the devices had only moved marginally. Co. A had tried to pass on cost increases, but in the depressed market this had proved impossible. Co. A had opted to keep running at full capacity with lower sales margins rather than risk a drop in sales at higher prices.

3.1 Analysis of the Current System

This is the first stage of the framework for the design and implementation of CIM in SMEs. At this stage, the company's product design and development and existing CIM technologies are analysed along with bottleneck situations arising during the design and implementation of CIM.

3.1.1 ProductDesignandDevelopment

Co. A has adopted CAD since 1986, and uses computer-aided modelling for testing the product performance parameters at the design stage. The drawing office holds a database of over 45 000 drawings in a network. There are still about 2000 old paper drawings in the department. Only the design staff in the drawing office have access to the CAD software in the network. Other staff of the company such as the sales managers, have to go through the design department to obtain a hard copy of the data. Once the drawing office receives an order with specific requirements, the staff extract the specific file and modify the design for creating a bill of materials for production.

According to the manufacturing processes manager, the particular electronic industry involved is mature and, there is no "real" revolution of products and hence there is no R&D department in the company. However, Co. A has developed many types of special electronic device over the last 40 years in response to customer needs. When a new specification is required by the customer, first of all, the drawing office produces the CAD drawing for computer-aided modelling at the design stage, this is for "testing" before "design". Then some samples are produced at the second time of modelling. This time, a half cut of samples are photographed, stored in the computer, and tested for specific performance criteria. According to the device development manager, this software package is very helpful in developing specific product requirements and is almost 95% efficient. The software is on the company network, but only staff within the technical department are authorised to access it.

3.1.2 ProductionPlanningandControl(PPC)

In Co. A, the duration of a time bucket is usually one week. However, the master production schedule (MPS) is based only on rough-cut capacity planning. Hence, the MPS and information of on-hand stock, manufacturing order status, order quantity, lead time and product structure are inputs to the MRP system. The output from the MRP system determines exactly how many of each item in each period are to be manufactured from the bill of materials (BOM). Then, the data are analysed and route cards for different batches are produced by computer for the production department.

In the system, the transfer of data from design to production gives rise to complicated transfer procedures and hence lost time. Geometry data already included in the CAD system and required for NC programming must be read from the drawing and re-entered. In general, the daily production of electronic devices is between two to three hundred. Bills of material consist mainly of several small mechanical components, and a few specified few other non-metallic materials. The route card is a document that is released from the MRP system. It goes to the production shopfloor and consists of information from the BOM and the workshop schedule, etc. The shopfloor has three divisions: the press shop, the non-metal shop, and the assembly divisions. They all have production responsibilities which are explained in the following.

The company purchases metals from suppliers in bulk sheets, which are formed into metal strips in the press shop. Then, they are cut into different sizes for different product performance requirements. The metal is processed mainly by manually controlled machines. There is a CNC mill and lathe but according to the press shop technician, there is not a great deal of production involving CNC. Similarly, the non-metal shop produces different types of tube from materials purchased from suppliers. The production of tubes is by automated machines. Together with the metallic components, the tubes, and other assembly parts purchased from suppliers, the devices are assembled manually, which is highly labour-intensive and requires skilled and semi-skilled workers. The products are then left overnight in a dryer at an operating temperature of around 200°C which evacuates the tube. After assembly, the products are transported to the quality inspection department. Each finished product is bar-coded individually by hand which gives the product an identification number before going into quality inspection. The batch is then ready for despatch to the customer.

3.1.3 OrganisationalCommunicationAnalysis

The company has been certified to ISO 9001 which covers documentation control, design control and the general management system. A communication network diagram of the company for sales orders is shown in Fig. 2.

It can be seen that it is a mix of paper and network systems. The company developed its existing CIM technology within the organisation as a SWAN network (LAN) which was introduced in 1993. In this, there is a link to customer orders through production management. Purchasing requirements and sales are linked to the account system. Once the customer has sent an order, the sales department inputs the data into the network. Then, the production department produces the specific product inventory file to purchase the necessary parts from suppliers. Then, MPS is created and a route card for each batch of the order is produced.

For price setting for the customer, an interrogation routine is as follows: the sales staff first provides an estimation for the customer; however, if the quotation is based on a batch of products in excess of 100, then the quotation must be authorised by the managing director through paper documents. Delivery time for the customer can be obtained via the Swan network-PPC, although occasionally unavoidable circumstances will render the process invalid, requiring it to be manually processed.

3.1.4 BottleneckSituations

1. *CAD/CAM link*. It is found that CIM is involved in separate islands of automation, such as CAD and CNC machining. The NC coding link between CAD and the CNC machine is not integrated, so the technician has to spend 60% of his time interpreting the NC coding. If this were fully integrated it could help to eliminate human error in programming the CNC machines.

2. *CASPIAN/SWAN network link*. From Fig. 2, it can be seen that the Caspian network is used only in the marketing and sales departments. This is another bottleneck for information flow. However, the Caspian network is preferable to the Swan network in these departments as its software is more suitable for storing customer details. Because of this, the marketing director has to meet the production director once a month for a marketing forecast meeting. The main purpose of this is to collect orders that the marketing department has forecast. Then, the information collected is manually interpreted and input to the Swan network for PPC. As most of the products are maketo-order, the information collected is mainly for production planning. However, if there is human error in interpreting the data and if there is an important change in sales forecasting which might affect the production planning dramatically, this could create problems as data input into the Swan network might not reflect reality. Hence, the production department may lose control over planning.

3. *Access of CAD drawing of products*. As mentioned above, apart from the drawing office staff, any other staff have to go through the drawing office to obtain a hard copy drawing. Time is lost in this way in responding to customers who make queries about the drawings through the sales staff.

4. *Access of computer-aided modelling of products*. Similarly, if a customer wants to discuss the specific performance requirements in detail together with the quotation for the products, he/she may have to go through two members of staff, one from the technical department and one from the sales department, to achieve an agreement.

5. *Sales order*. Because of the communication links within the organisation mentioned, sometimes a good definition of discrete product order from the sales department is hard to achieve. This is due to the missing communication link between the sales department, MPS and the

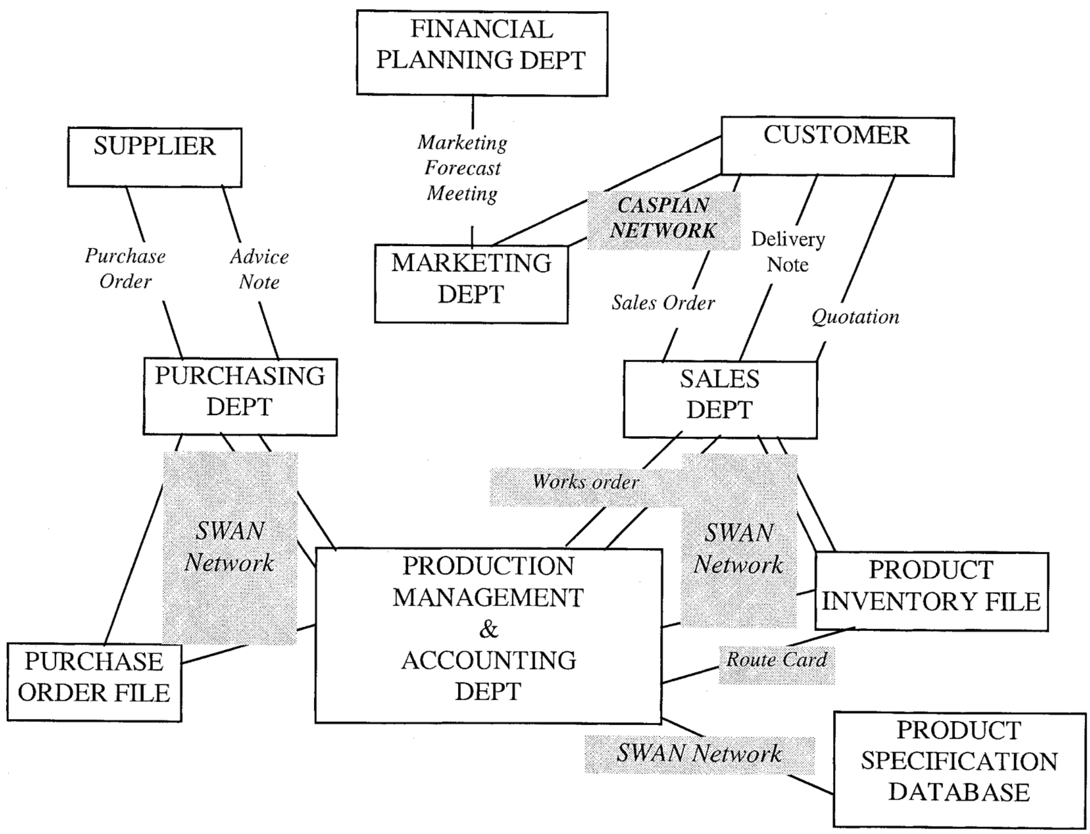

Fig. 2. Communication links within Co. A.

product specification database in which are stored important test parameters of products, as shown in Fig. 2. If they were linked together, this could reduce much misunderstanding.

6. Organisational behaviour towards the SWAN network. Although most of the staff interviewed are fairly happy with the system, as the network is still relatively new to the staff, some of them lack problems in adjusting to this new environment. They have doubts about the real purpose of this new working environment, which is aimed at reducing WIP. It seems to them that the major problem with the SWAN network is the cumbersome routine of data input and of obtaining the right information from it.

3.2 Design of the CIM System

In this section, the design of the CIM system is discussed and the company's business, manufacturing strategies, benefits of CIM implementation, network architecture and PPC system are explained, and the workforce requirements, plan for staff training and design organisational interfaces for the network architecture are discussed. The following business and manufacturing objectives should be considered in the design of the CIM system:

1. To maintain the consistency of the quality of products.
2. To deliver products on time.
3. To offer more products to customers.
4. To design products which will improve performance.
5. To design electronic devices which can be made wholly onthe shopfloor.

3.2.1 TheAimsandBenefitsoftheImplementationof CIM

1. *To integrate the business information and manufacturing planning*. As said earlier, it is often found that information from the company network does not reflect reality. For this reason, it is essential to link the sales department and product specification database to increase the information flow for customer orders to provide a more realistic manufacturing plan and hence increase flexibility.
2. *To increase the amount of data input and data collection into and from the network*. Similarly, CAD and computeraided modelling software is only available to authorised staff. In order to use the network, CAD files and modelling files should be made readily available to any other staff on a read only memory (ROM) basis from the network to reduce the complications of communication.
3. *To reduce WIP*. As mentioned earlier, there is a missing link between CAD and CAM. A press shop technician has to spend 60% of his time interpreting the NC code. If software is linked between CAD and CAM to convert NC coding, this 60% of labour time could be reduced greatly. It will also reduce human error, making communication easier for achieving and assuring product quality.
4. *To increase the use of bar-coding at the production stage*. Currently, the bar-coding system is used only after the assembly of the electronic devices. Information in the route cards consists only of product delivery time, and the quantities and the quality specifications and work schedules. However, random events may occur on the shop floor, such as failure of machine tools, shortage of supplies and absence of workers. These random events force changes in the schedules. Implementing a bar-coding system for WIP and distribution tracking would maintain efficient shop-floor control.

3.2.2 LevelofComplexity

1. *PPC*. At the moment, production planning is based on MRP. According to the manufacturing processes manager, no change to other systems such as JIT is anticipated. This is due to the nature of the products in which a few orders can be combined to produce common requirements of parts to optimise production, whereas JIT would lose the discrete identity of a product. A number of barriers to implementing JIT are:

   1 Frequent changes in production planning.

   2. Inaccurate forecasting procedures resulting in under/ over-forecasting.
   3. Equipment failures creating capacity problems.
2. *Automated assembly*. The assembly line of the products is highly labour intensive. Introducing robotics for assembly will not be considered by Co. A for the foreseeable future, owing to the large investment that would be involved. According to the company, manual assembly of the electronic devices is efficient for the current standard. However, Co. A's biggest competitor in Japan has recently adopted robotics for assembly, therefore, if Co. A aims to increase competitiveness, it should consider implementing automated assembly in its long-term planning.
3. *Automated material handling and storage*. On the shopfloor, materials such as work in progress and subassemblies are transported by two-wheeled hand trucks. Raw materials and finished assemblies are loaded and unloaded by forklift trucks. Owing to the constraints of shop-floor space and the existing plant layout, an automated material handling system such as conveyors is currently impossible to install in the company. However, Co. A regards this as one of the longterm planning issues.

3.2.3 EstimatetheAbilityofWorkforceandPlanfor StaffTraining

Owing to the reasons mentioned, Co. A has set up a technical support team, the "call" team. This team is to be

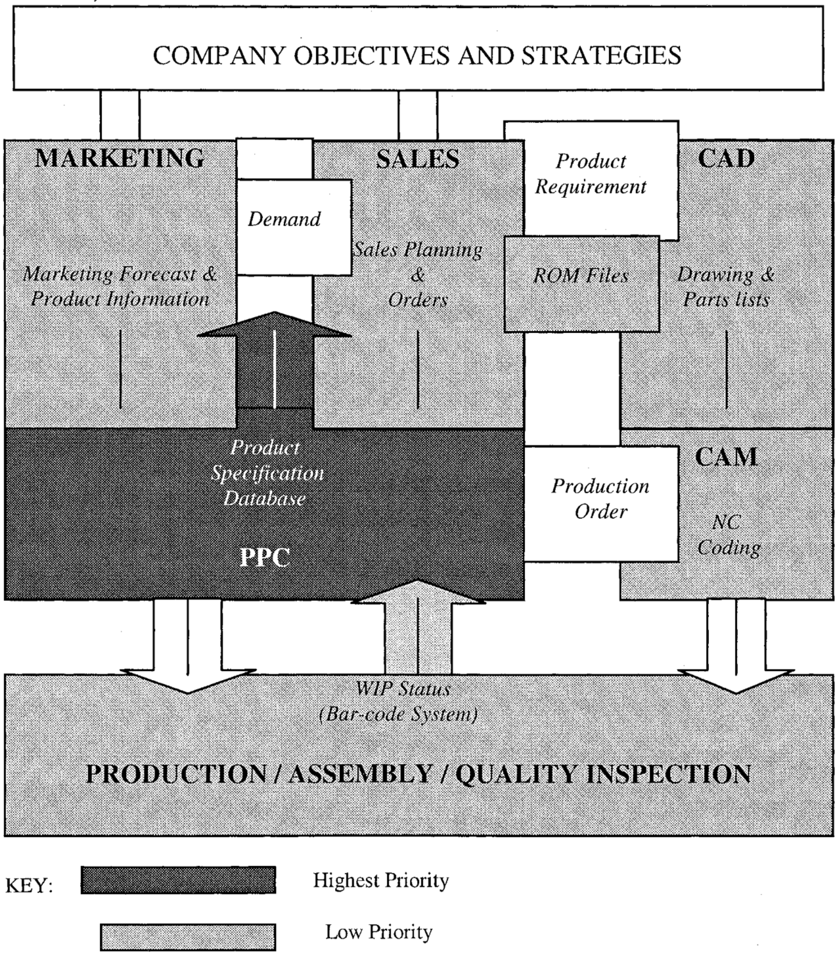

Fig. 3. New network architecture.

called when help is required by any other staff in the company who have IT problems with the network. Its members consist of the production engineer, sales administrative, business administrative, the finance director, the IT manager and the manufacturing processes manager. The purpose of the team is to allay the doubts and the dissatisfaction of the staff. Currently, some of the staff still receive two hours of training per day on a one to one basis.

3.2.4 DesignOrganisationalInterfacesforNetwork Architecture

As a summary of all the above subsections, a new network architecture is designed as shown in Fig. 3. Taking the proposals for solutions contained in the above subsections as a starting point, CIM subfunctions are defined in the above diagram. This is a tailor-made network architecture, showing the CIM building blocks and a great deal of attention should be given to those requirement profiles specified here. This is an integration path which Co. A should follow, integrating the product specifications database with sales, marketing forecast and PPC, ROM CAD files on the network, and a bar-code system with production and assembly. These are the highest priorities.

3.3 Implementation of the CIM System

This section is concerned with the implementation of a CIM system. At this stage, the availability of the CIM modules, the implementation of the organisational interfaces for network architecture and the training of staff is discussed. Also, time critical, step by step integration of the CIM working environment with trained and skilled staff is briefly discussed.

3.3.1 BuyNecessaryCIMModules

1. *Sales planning/product specification database*. Since the marketing and sales departments are both in the Caspian network, Co. A is advised to consult the vendors to discuss the possibility of linking Caspian and SWAN together.
2. *Bar-coding system*. The use of current bar-coding equipment in the company is a problem. Current working temperatures means that bar codes cannot be applied, as they would melt at these temperatures. Having a bar-coding system, which could withstand these temperatures, would be great advantage.

3. *CAD/CAM link*. To rectify the problem of generating NC coding from CAD drawings, there are many software packages available in the market, one of which is the Mastercam. Drawings can be produced using this software, and then the NC coding will be generated automatically for the appropriate machines if an appropriate post processor is installed and linked into the CNC machine, the CNC machine will be automatically programmed to carry out the task.

3.3.2 ImplementtheOrganisationalInterfacesfor NetworkArchitecture

The SWAN system was introduced three years ago, and has been implemented incrementally to take over the old system. For instance, it was first introduced in the finance department. During the second year, it was introduced in the sales department and then it was introduced to the production department. Attempts should be made to reduce the overall customer order duration by considering linking the sales department and product specification database as previously mentioned. Furthermore, if CAD/CAM software was installed, there would be a link between the drawing department and the press shop in the company.

3.3.3 TrainingofStaff

If the company accepts the idea of purchasing CAD/CAM software, a training course should be provided for the relevant staff, such as the drawing technician and the NC programmer. However, according to the author's experience, most of the learning would be gained through experience, as it is essentially "learning by experience" software. Furthermore, some of the sales and marketing staff might have to be retrained if the link between Caspian and SWAN is successful.

3.3.4 StepbyStepTimeCriticalIntegrationintothe CIMWorkingEnvironmentwithTrainedandSkilled Staff

For any SME to achieve a totally implemented CIM system, there is a long-term integration issue. The development of CIM differs in terms of the time horizons involved and details required. On the one hand, this is due to the limited amount of investment and number of skilled staff available. On the other hand, it is due to the level of organisational change that might be involved. Having reviewed the company current situation, it is suggested that the company is now at the stage of "short-term planning". If the company takes the actions to rectify the bottleneck situations mentioned earlier, it would hopelessly achieve the system's objectives in the long-term, as shown in Table 1.

4. Summary of Findings and Recommendations

The following are some of the findings and recommendations for the design and implementation of CIM in SMEs.

1. Having applied the framework onto Co. A, the evidencegathered from the case study has suggested some specific actions for the company to take in order to use its CIM technologies and to increase its competitiveness in the market. Within the practical framework developed, it is shown that there is a close theoretical connection between software and hardware adoption, and learning and business development. It has also identified the architecture, the layout of different components, the communications between those elements, the software requirements and the underlying organisational infrastructure requirements. Moreover, it has identified the priority areas and the overall preferred sequence of implementation. Managers within the organisation can now see the company future in a different light.

   These suggestions would improve the links in the information flow in the company. Regular reviews of CIM adoption progress using this framework could help to identify the inefficiencies, and consequently considerable

Time horizon	CIM strategic development and operational issues	Organisational responsibility
Day-to-day	Individual activities with islands of automation. Meeting delivery schedules once the information has been provided.	Shop floor
Short-term	Management initiative with accelerated links of data and resources, such as links between sales order and product specification database. Extensive sharing of data through network, such as ROM CAD files available on the network.	Horizontal integration control
Aggregate planning	Management controls designed within the long-term plan. Consist of establishing feasible medium-term (up to two years) plans to meet agreed integration levels, such as link between marketing forecast and PPC and introducing barcoding system at the production stage.	Management strategic control
Long-term	System momentum. Long-term (five years or more) capacity planning. Involves important strategic issues such as introducing automated material transport system and automated assembly. Also skilled and trained staff anticipate technical developments. Common distribution network fully developed.	Whole organisation

Table 1. Time critical integration of CIM in Co. A.

benefits should accrue when integration becomes technically feasible.

2. Having realised the applicability of this framework, it isfound that this framework applied to Co. A is a “bottomup” approach. Individual areas and line managers have to undertake CIM implementation initially based on their local needs. Then, cross-functional integration among various business functions becomes a later issue. Owing to the financial and organisational limitations of SMEs, this approach is inevitable, if islands of automation already exist in the company. This is found to be particularly true in Co. A, in which, for instance, the sales and marketing departments favour the Caspian network for their own needs. The Caspian network has become disintegrated from the SWAN network which has created a huge problem to the company in terms of flexibility and use of the network. The key issue here is that vendors usually develop and deploy the standards that allow decentralised development of specific CIM components. This has misled companies into implementing islands of automation with localised benefits, which produce systems that tend to become more and more specialised over time, and the integration of these system often remains neglected.
3. Companies such as Co. A have employed dedicated hardware, LAN protocols, operating systems, product specification data dictionaries and particularly data structures in ways that make later plant integration even more costly. From the case study, it is apparent that the implementation of CIM is being motivated and justified on traditional grounds, such as reducing cost and improving WIP. However, motivations such as increasing assembly automation, and the need to learn about CIM are not major factors in the company. The company claimed individual operational improvement as the major criterion for selecting the network, but the feasibility of network communications within the whole organisation was not viewed as an important issue initially. For instance, one manager admitted, “the two different systems developed will take years to communicate with each other”.
4. There are many potential pitfalls in the area of CIM planningand implementation that must be avoided. The first is to ignore CIM or underestimate its impact. This kind of management myopia is frequently found in conventional SMEs. The second pitfall is to lose touch with the rapid pace of CIM developments. What was true two years ago about the availability, cost and performance of a given system, may be totally outdated today [13]. In general, this practical framework can be perceived as being driven by supply push or by demand pull. However, it could be inefficient if supply push is retarded by market fragmentation caused by the idiosyncratic needs of diverse types of user and by a general lack of understanding of SME needs by suppliers, or if demand pull is hampered by a lack of awareness by SMEs of the possibilities that CIM affords them and their lack of ability to address the CIM adoption process.
5. Changes in the company usually result from strategic andoperational decisions such as continuous improvement, changes in product design, changes in product volume, and evolving business processes. Decisions must be made about how to deal with each change. Such decision making may involve many individuals in the company. Once decisions have been made, the impact of the change must be communicated to all those affected in the company. Tools and technologies for group decision making and communication among employees, customers, and suppliers will be invaluable for achieving implementation of CIM.
6. If we examine the current information system at Co. A,there is lack of attention to information technology by the capacity adjustment and conveyance control departments. There is a need to concentrate on those departments for running the system smoothly. Also, little attention is given to order control (sales), capacity requirement planning, operation data collection, and NC programming. These sections are the backbone of the plan for the survival of the company. Therefore, due attention is required to these sections to compete in the global market. Similarly, little attention is given to information technology in the purchase of equipment, assembly control and maintenance.

5. Conclusions

In this paper, the concept of design and implementation of CIM in SMEs has been discussed and analysed for better understanding of the concept. The literature available on the design and implementation of CIM has been reviewed with the objective of developing CIM in a real-world application. It has been possible to propose a time-critical framework for the design and implementation of CIM in SMEs, which is based on the reviews and analyses of approaches available on CIM. It has provided a refined holistic model for identifying inefficiencies in the overall process of designing, developing and implementing CIM in an SME. In the near future, it seems likely that the gap between the theory and practice of CIM will be bridged. The research shows that the development of shared CIM systems is highly likely, having demonstrated the pros and cons. The tools available should be tailored for the design and implementation of CIM in their total integrated form. At present, there is a lack of integration of shared CIM systems, which can only be overcome by the examination of the various approaches and methodologies from each CIM nucleus, that could possibly affect the efficiency of a shared CIM system.

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